

The Annuity Puzzle Reconsidered: A Conditional Framework for Household Heterogeneity and Institutional Variation

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Abstract

Whether observed take-up of voluntary life annuities — roughly 3 to 6 percent among single retirees in the US and UK — reflects rational household decision-making or welfare-relevant under-annuitization depends on intra-household longevity-risk sharing, intergenerational transfer motives, long-term care and Medicaid exposure, pre-existing public and defined-benefit annuitization, and the institutional default architecture facing the household at retirement. This Perspectives article asks when low annuity demand is well-explained by household heterogeneity and institutional frictions, and where it remains a welfare-relevant puzzle. Calibrated lifecycle models converge on several demand-reducing channels but disagree about their joint welfare relevance, especially for households with high financial wealth, low bequest motives, good health, and limited pre-annuitized income. The article organizes the evidence into a conditional framework by product or institutional setting and household type. For retail single-premium immediate annuities, observed low take-up is well-explained for many lower-wealth households, married couples with spousal pooling, and households with strong bequest motives; the welfare answer remains contested for high-wealth childless retirees in good health. Across deferred income annuities, pension payout defaults, Social Security claiming, and mandatory annuitization regimes, the welfare question is either contested or under-examined. The resulting research agenda shifts attention from the aggregate annuity puzzle to heterogeneous welfare effects, retirement-income product design, and household-facing policy and default architecture.

Keywords: annuity puzzle; life-cycle theory; longevity risk; bequest motives; household heterogeneity; retirement income

JEL Classification: D15, D81, D91, G22, H55, J26

1. Introduction

Yaari (1965) showed that under specific conditions, a rational consumer facing an uncertain lifetime, with no bequest motive and access to actuarially fair life annuities, should allocate all wealth to annuity contracts. The gap between this prediction and observed behavior has been called the “annuity puzzle”: voluntary immediate annuity ownership rates run approximately 3 to 6 percent of single retirees in the US and UK (Lockwood 2012; Inkmann, Lopes, and Michaelides 2011), with comparably low take-up of lifetime-income products in other developed economies such as Australia (Australian Treasury 2023). The welfare stakes are real because annuities pool longevity risk in ways unavailable through conventional saving, so any unexplained under-utilization may reflect market failure, behavioral frictions, omitted household motives, or institutional and product-design choices.

Research over six decades has identified substantial channels that reduce predicted demand below the Yaari benchmark. Social Security and defined-benefit (DB) pensions already annuitize 50 to 75 percent of total wealth for the median retiree (Lockwood 2012; Dushi and Webb 2004). Bequest motives are empirically prevalent and quantitatively important for intergenerational transfers (Kopczuk and Lupton 2007; Lockwood 2012). Married couples capture substantial fractions of fair-market welfare gains through intra-household longevity-risk sharing (Kotlikoff and Spivak 1981; Brown and Poterba 2000). Health expenditure risk correlated with stochastic mortality, and means-tested public coverage of long-term care, alter the value of annuitized income in late-life states (Reichling and Smetters 2015; Ameriks, Caplin, Laufer, and Van Nieuwerburgh 2011). Pricing loads of 15 to 20 percent against population mortality (Mitchell, Poterba, Warshawsky, and Brown 1999; Wettstein, Munnell, Hou, and Gok 2021) and product-market frictions further suppress demand.

These channels do not yet jointly explain low voluntary demand within a unified calibrated model. Pashchenko’s (2013) structural model incorporating multiple impediments still overpredicts single-premium immediate annuity (SPIA) participation by a factor of four; her tightened institutional specification reduces predicted participation to under 8 percent without invoking new economic channels. Peijnenburg, Nijman, and Werker (2016) find full annuitization approximately optimal in a calibrated model where agents save out of annuity income, though their European setting assumes negligible out-of-pocket health costs. Inkmann, Lopes, and Michaelides (2011), using UK data, find that reasonable preference parameters match observed low participation. Which households are well-explained by existing channels, and which represent welfare-relevant under-annuitization, depends on which calibration is operative.

This paper organizes the evidence into a conditional framework by product or institutional setting and household type. The framework distinguishes retail single-premium immediate annuities, deferred income annuities and qualified longevity annuity contracts (QLACs), pension payout choices, Social Security claiming, and mandatory or default regimes. Each product/setting $\times$ household-type cell is classified as well-explained (the established channels account for observed low demand for the modal household in the cell), contested (calibrated models disagree on welfare-relevant magnitudes), or under-examined (the welfare question remains largely unaddressed within the canonical puzzle literature). The classification

separates positive explanation from welfare judgment: a well-explained cell may still contain individual households who would benefit from annuitization, and welfare gains may be larger under improved product menus. The framework’s contribution is to direct attention from the aggregate annuity puzzle toward the household-economics topics that determine the welfare answer: spousal pooling, intergenerational transfers, long-term care and Medicaid exposure, retirement-income product design, and default architecture.

Prior reviews of the annuity puzzle have organized the literature by behavioral channel (Benartzi, Previtro, and Thaler 2011), by structural and behavioral mechanism (Brown 2007), or as systematic syntheses of annuitization decisions and product-market frictions (Alexandrova and Gatzert 2019; Lambregts and Schut 2020). Relative to these reviews, the contribution here is to reorganize the evidence by household type and product or institutional setting jointly, separating positive explanations of low demand from welfare judgments and identifying which cells of the resulting framework remain genuinely open.

The literature surveyed spans the six decades from Yaari’s original 1965 framework to recent work in 2025. Studies were identified through keyword searches in EconLit, Google Scholar, RePEc/IDEAS, and NBER Working Papers, supplemented by citation chasing from canonical anchors. The review is integrative rather than systematic: included studies are those that develop canonical theory, calibrate lifecycle models with direct welfare implications for one or more framework cells, provide empirical or experimental evidence on a specific channel, or document institutional variation in annuitization mandates and defaults. The selection is focused on contributions that bear directly on the conditional framework rather than on an exhaustive enumeration of the broader retirement-finance literature.

2. The Yaari benchmark

Yaari (1965) proved that a consumer facing an uncertain lifetime, with no bequest motive and access to actuarially fair life annuities, should allocate all wealth to annuity contracts. With fair annuities, the consumer earns a return equal to the bond rate plus the mortality hazard rate, the “mortality credit,” financed by redistributing wealth from those who die to survivors. The annuity stochastically dominates the bond for any consumer who derives no utility from wealth at death, and full annuitization is optimal.

The result rests on a conjunction of nine assumptions, summarized in Table 1, none of which generally holds without qualification as a descriptive matter. Expected utility, additive separability, and absence of bequest motives are explicit in Yaari’s setup. Actuarially fair pricing, a known and fixed survival function, and exponential discounting are standard in subsequent implementations. The remaining three (absence of pre-existing annuitized income, absence of expenditure shocks beyond mortality, and state-independent utility) are implicit features of Yaari’s modeling environment that subsequent research identified as empirically load-bearing.

Table 1. Nine Assumptions of the Yaari Benchmark and Their Empirical Status

No.	Assumption	Status in Yaari (1965)	Key Paper(s) Identifying Importance	Empirical Status
1	Expected utility (von Neumann-Morgenstern)	Explicit	Hu and Scott (2007); Brown et al. (2008)	Strong evidence against: systematic violations including loss aversion, probability distortion, and framing dependence
2	Additively separable utility	Explicit	Bommier and Le Grand (2014)	Theoretical challenges from non-additive lifetime risk preferences that interact with annuity demand
3	No bequest motive	Explicit	Lockwood (2012); Kopczuk and Lupton (2007)	Strong evidence against: approximately 75% of elderly singles estimated to have bequest motives (Kopczuk and Lupton, 2007); 67% of those with children report leaving inheritance as important (Lockwood, 2012)
4	Actuarially fair pricing	Explicit	Mitchell et al. (1999)	Not satisfied for population mortality: MWR 0.80-0.85; closer on annuitant mortality: MWR 0.90-0.94
5	Known, fixed survival function	Explicit	Reichling and Smetters (2015); O'Dea and Sturrock (2023)	Strong evidence against: mortality is stochastic and correlated with health expenditures; individuals systematically underestimate survival probabilities
6	Exponential discounting	Standard in applications (Yaari uses a general discount function)	Laibson (1997)	Contested: evidence of present bias and quasi-hyperbolic discounting
7	No pre-existing annuitized income	Implicit	Dushi and Webb (2004); Bernheim (1991)	Strong evidence against: Social Security and pensions constitute 50-75% of retiree wealth
8	No expenditure shocks beyond mortality	Implicit	De Nardi, French, and Jones (2010); Reichling and Smetters (2015)	Strong evidence against: stochastic medical costs (combined out-of-pocket and Medicaid) with mean rising from \$5,100 at age 70 to \$29,700 at age 100 and 95th percentile reaching \$111,200/year at age 100 (Jones et al., 2018; in 2014 dollars)
9	State-independent utility	Implicit (in the single-good consumption specification)	Finkelstein, Luttmer, and Notowidigdo (2013); Brown, Goda, and McGarry (2016)	Contested in direction: strong evidence of state dependence, but sign varies: FLN (2013) find marginal utility declines with health deterioration; Lillard and Weiss (1997) find the opposite; Brown et al. (2016) document heterogeneity across health conditions

Note: Assumptions 1-5 are explicit conditions of Yaari's proof, with 1 and 2 relaxable under complete markets (Davidoff, Brown, and Diamond, 2005). Assumption 6 is standard in the applied literature but not an explicit condition of Yaari's general formulation. Assumptions 7-9 are features of Yaari's modeling environment that subsequent research identified as empirically load-bearing. For most assumptions, relaxation reduces predicted annuitization; assumption 9 (state-independent utility) has directionally contested effects, and assumption 6 (exponential discounting) can in principle move predicted demand in either direction depending on calibration. The framework in Section 7 treats the direction-ambiguous cases as part of the contested cells rather than as uniform demand-reducing channels.

Davidoff, Brown, and Diamond (2005) generalized the result by showing that with no bequest motive, complete markets, and a return premium net of costs, full annuitization is optimal regardless of utility function. This relaxes the preference assumptions but shifts the load-bearing assumption to complete markets, a condition real annuity markets, dominated by nominal fixed products, do not satisfy. Davidoff, Brown, and Diamond also established a weaker result: some positive level of annuitization is welfare-improving under broader conditions, consistent with annuities not being valueless even when full annuitization is not predicted. Relaxing many of the empirically central assumptions reduces optimal annuitization; along several dimensions, calibrated models predict demand at or near zero, while assumption 9 (state-independent utility) and assumption 6 (exponential discounting) can have directionally ambiguous effects depending on calibration. Sections 3 through 6 review the channels through which these assumption failures translate into reduced predicted demand; Section 7 organizes those channels by product and household type.

3. Pre-existing annuitization and bequest motives

Two channels, considered jointly, substantially narrow the gap between the Yaari prediction and observed behavior: the annuitized wealth that households already possess through Social Security and defined-benefit pensions, and the prevalence of bequest motives.

3.1 The pre-annuitization baseline

The standard documentation of the annuity puzzle compares private annuity wealth to financial wealth, understating the degree to which retirees are already insured against longevity risk. The relevant measure is the share of total retirement resources already in life-contingent form, including Social Security and defined-benefit pensions, both of which are lifelong and largely non-commutable. Lockwood (2012) reports that households in the second and third deciles of the wealth distribution have approximately 75 percent of total wealth already annuitized; at the median, approximately 67 percent; at the 80th to 90th percentile, roughly half. Only in the top decile does the share fall to approximately 33 percent. Poterba, Venti, and Wise (2011) find that Social Security and defined-benefit wealth constitute approximately 46 percent of mean household wealth at retirement. The pattern is inversely related to financial wealth because Social Security benefits are progressive. Dushi and Webb (2004) confirm that much of the failure of the average currently retired household to annuitize can be attributed to the high proportions of cohort wealth already pre-annuitized. These estimates reflect cohorts with substantial DB pension coverage, currently approximately 14 percent of private-industry workers in the United States (Bureau of Labor Statistics, 2025). As DB coverage declines, the pre-annuitization share for future retirees will rest principally on Social Security alone, making the channel cohort-dependent.

3.2 Bequest motives

Yaari's (1965) theorem assumes the consumer places zero value on wealth remaining at death. Relaxing this assumption moves theoretical predictions much closer to observed behavior. Kopczuk and Lupton (2007) estimate, using structural estimation of a lifecycle consumption model for elderly singles, that approximately 75 percent have a bequest motive. Among those with children, 67 percent report that leaving an inheritance is important; among those without children, 55 percent express a desire to leave bequests (Lockwood, 2012). The desire to provide for descendants is anthropologically widespread, consistent with Hamilton's (1964) theory of inclusive fitness and documented in cross-cultural human life-history research (Kaplan, Hill, Lancaster, and Hurtado 2000; Hooper, Gurven, Winking, and Kaplan, 2015).

The quantitative impact on annuity demand is large. Lockwood (2012) finds that without bequest motives, his lifecycle model predicts 61 percent SPIA ownership among single retirees aged 65 to 69, with willingness to pay of 25.3 percent of non-annuity wealth for actuarially fair contracts. With moderate bequest motives, willingness to pay collapses to 3.7 percent; with annuities carrying a 10 percent load, it falls to zero. Adding a fixed purchase cost of approximately \$2,000 reduces predicted ownership to 4 to 5 percent, aligned with the observed 3.6 percent. The interaction with annuity pricing loads strengthens the result:

bequest motives and realistic loads are multiplicative in suppressing demand. Kotlikoff and Spivak (1981) show that married couples can achieve approximately 70 percent of the welfare gains from fair annuity markets through intra-household risk sharing, further qualifying the case for additional voluntary annuitization, though this single-calibration estimate predates modern treatments of divorce risk and late-life household dissolution. For US taxable financial assets, inherited holdings receive a step-up in cost basis that eliminates unrealized capital gains tax, while annuity payouts are taxed as ordinary income. The comparison differs for qualified retirement accounts, nonqualified annuities with exclusion ratios, and other jurisdictions, but the directional effect generally favors bequeathable financial wealth over annuitized payouts in US taxable accounts.

4. Health expenditure risk and stochastic mortality

Retirees face medical and long-term care costs that are large, unpredictable, and concentrated in states where liquidity is most valuable. Annuities convert liquid wealth into an illiquid income stream and exacerbate this tension under specific correlation structures.

4.1 Liquidity and the right tail of medical risk

Health expenditure risk in retirement is concentrated in the right tail. Jones, De Nardi, French, McGee, and Kirschner (2018) estimate that combined out-of-pocket and Medicaid expenditures rise from approximately \$5,100 at age 70 to \$29,700 by age 100, with the 95th percentile of combined annual expenditures reaching approximately \$111,200 at age 100. The dominant uninsured tail risk is long-term care, which Medicare covers only for short post-acute episodes. The median annual cost of a semi-private nursing facility room was \$111,325 in 2024 (Genworth, 2024). These costs are lumpy, unpredictable, and positively correlated with deteriorating health.

4.2 Precautionary saving and public care aversion

De Nardi, French, and Jones (2010) show that medical expense risk drives substantial precautionary wealth accumulation, particularly among higher-income elderly singles. Ameriks, Caplin, Laufer, and Van Nieuwerburgh (2011), using strategic survey methods to separate bequest motives from public care aversion (the desire to avoid Medicaid-funded care), find both motives are economically significant. For some households, liquidity retention may reflect a preference to maintain capacity to self-fund high-quality care rather than a failure to understand longevity insurance. Annuitizing converts liquid wealth into an irrevocable payment stream, reducing the option value of holding spendable assets.

4.3 Stochastic mortality correlated with health costs

Reichling and Smetters (2015) provide a particularly strong demand-suppressing formulation. They allow health shocks to alter survival probabilities stochastically: a negative health shock simultaneously increases the marginal utility of wealth (through elevated medical

costs) and reduces the remaining present value of annuity payments (through shortened life expectancy). The annuity becomes “valuation-risky” because a health shock generates implicit depreciation in annuitized wealth exactly when liquid resources are most valuable. When this health-mortality-cost correlation is calibrated to realistic magnitudes, optimal annuity demand falls to zero or even becomes negative. This result operates even under actuarially fair pricing and even absent bequest motives, conditions under which the Yaari result and its Davidoff, Brown, and Diamond (2005) generalization both predict full annuitization.

5. Behavioral models

The preceding sections relaxed the Yaari framework’s assumptions within the expected utility paradigm. This section asks what happens when the behavioral model itself is replaced. Under standard behavioral parameterizations, predicted annuity demand can fall to zero.

5.1 Framing effects

Brown, Kling, Mullainathan, and Wrobel (2008) demonstrate that description invariance, a foundational expected-utility axiom, fails sharply in the annuity context. Approximately 72 percent of respondents preferred the annuity when framed as consumption insurance, versus only 21 percent when framed as an investment. The 51-percentage-point swing is strong evidence against description invariance in stated annuitization preferences. Consistent directional evidence appears in subsequent designs (Agnew et al., 2008; Beshears et al., 2014). A factor-of-three swing in stated preference from framing alone is difficult to reconcile with the stable underlying preferences that expected utility theory presupposes.

5.2 Prospect theory and predicted demand

Hu and Scott (2007) apply cumulative prospect theory formally to the annuity decision. Their central measure is a reservation price: the maximum fraction of actuarially fair cost that a CPT agent would pay for an annuity. Under the full CPT specification with standard Tversky-Kahneman parameters and narrow framing, reservation prices for a 65-year-old male fall to approximately 0.77, implying zero demand at fair or loaded prices. Two CPT mechanisms drive this result. Loss aversion weights early-death losses at 2.25 times equivalent longevity gains, and probability weighting overweights the small probability of dying soon after purchase. The result is sensitive to reference-point specification and to the loss aversion coefficient. Hu and Scott’s own Table 1 shows that far-deferred longevity annuities can be attractive under full CPT: a 65-year-old’s reservation price for an annuity beginning payouts at age 95 reaches 1.20, a result relevant to the deferred income annuity puzzle in Section 7.2.

5.3 Survival pessimism

O’Dea and Sturrock (2023) document that individuals systematically underestimate their own survival probabilities, by approximately 25 to 30 percentage points among those in their early sixties estimating survival to age 75, and by 15 to 20 percentage points among those in their

late sixties estimating survival to age 85. In a calibrated lifecycle model, survival pessimism markedly reduces perceived annuity value: when individuals believe they are unlikely to survive to advanced ages, the expected present value of annuity payments, as perceived by the decision-maker, falls well below the premium.

5.4 Bounded rationality

The annuity purchase requires estimating a remaining lifespan distribution, evaluating actuarial fairness, accounting for inflation erosion over decades, comparing across product structures, and assessing insurer solvency. These tasks are demanding relative to the financial literacy documented for retirement-age adults (Lusardi and Mitchell, 2011). Complexity interacts with status-quo bias. In many retail settings, the status quo is to retain the lump sum, and there is no institutional nudge toward annuitization comparable to the automatic enrollment that transformed 401(k) participation rates (Madrian and Shea, 2001).

6. Supply-side frictions and cross-cultural variation

The annuity in Yaari’s model assumes actuarial fairness, inflation indexation, default-free pricing, and frictionless markets. None of these holds in retail annuity markets, as this section reviews.

6.1 Pricing and money’s worth

The foundational empirical study of annuity pricing is Mitchell, Poterba, Warshawsky, and Brown (1999). Using actual quotes from the US individual annuity market, they report money’s worth ratios of roughly 0.80 to 0.85 against population mortality tables and roughly 0.90 to 0.94 against annuitant mortality tables. The gap between population-table and annuitant-table MWRs reflects the longer lives of voluntary annuity purchasers. The remaining shortfall reflects insurer expenses, statutory reserves, distribution costs, taxes, and profit. Wettstein, Munnell, Hou, and Gok (2021) found this pricing wedge stable over 2001 to 2019. Finkelstein and Poterba (2002), using the UK’s dual voluntary-compulsory market structure, demonstrated that adverse selection is roughly twice as severe in voluntary markets as in compulsory markets. Mitchell et al. (1999) note that adverse selection alone is unlikely to fully account for limited voluntary demand.

6.2 Other supply-side frictions

Most annuities sold are nominal: at 3 percent inflation, a nominal annuity loses roughly 45 percent of purchasing power by year 20. Inflation-indexed annuities are rare, if available at all, with initial payouts approximately 30 percent lower than comparable nominal products (Brown, Mitchell, and Poterba, 2002). A fixed annuity is, from an economic perspective, a long-term illiquid claim on an insurance company’s general account. State guaranty associations provide backstops with coverage typically set at \$250,000 in present value of annuity benefits (NOLHGA, n.d.). Total US annuity sales reached a record \$434.1 billion

in 2024 (LIMRA, 2025), but SPIAs accounted for only \$13.6 billion and DIAs \$4.9 billion: less than 5 percent of total annuity sales. Most consumers purchasing annuity products are buying accumulation or principal-protection vehicles, although guaranteed lifetime withdrawal benefit (GLWB) riders attached to variable and fixed indexed annuities do provide longevity insurance while preserving residual liquidity, complicating the simple SPIA-DIA-only picture of longevity-product take-up. Distribution channel conflicts further suppress demand: assets-under-management advisors have a direct financial interest in discouraging annuitization, while commission-based advisors face opposite incentives that favor complex, high-commission products. Strategic price complexity in retail financial markets further inhibits comparison shopping (Carlin, 2009).

6.3 Cross-cultural institutional variation

Cross-cultural evidence provides an institutional natural-experiment frame. Table 2 summarizes the major cases.

Table 2. Cross-cultural evidence on annuity demand

Context	Policy variation	Measure	Outcome	Source
United Kingdom (2015)	Removed mandatory annuitization (“pension freedoms”); simultaneously introduced flexi-access drawdown	Contract volume (flow)	Quarterly sales fell from 74,092 to 18,243 (~75% decline; ABI data reported in FCA, 2018); annual sales rebounded to 89,600 contracts by calendar year 2024 (ABI, 2025) driven by higher interest rates; confounders include RDR advice market reform and near-zero rate environment	FCA (2018); FCA (2019); ABI (2025)
TIAA-CREF, United States (2000–2018)	Declining institutional norms favoring annuitization	First-time income choice	Fraction of first-time retirement income recipients selecting a life-contingent annuitized payout declined from 61 percent to 18 percent; among those who deferred income until RMD age, the share fell from 41 percent to approximately 6 percent	Brown, Poterba, and Richardson (2025)
Australia (ongoing)	No annuitization mandate in compulsory superannuation system; ABP drawdown as dominant default	Asset stock share	Lifetime annuity products constitute a negligible fraction of approximately AUD 3.6 trillion in superannuation assets; Retirement Income Covenant (effective July 2022) mandates trustees formulate longevity-risk strategies	Australian Treasury (2023)
Singapore (2009/2013)	CPF LIFE introduced voluntarily in 2009; mandatory from 2013 for members with sufficient balances	Compliance rate	High compliance under mandatory status; broader CPF policy framework generated public debate	Fong, Mitchell, and Koh (2011); CPF Board
Chile (1981–present)	AFP system with choice between programmed withdrawal and immediate annuity at retirement	Share of new retirees choosing annuity (flow)	Over 60% of eligible retirees purchase private annuities; annuity take-up varies by cohort, wealth, and marital status. Early retirement rules and minimum pension thresholds substantially constrain the choice set, so high take-up partly reflects institutional design rather than purely voluntary choice	Illanes and Padi (2019)
Netherlands (ongoing)	Quasi-mandatory occupational pensions covering ~90% of workforce; limited lump-sum access	Institutional coverage rate	High annuitization by design; occupational pensions (historically DB and collective arrangements) provide lifelong income; system operates collectively rather than through individual retail purchase. The Future of Pensions Act (effective 1 July 2023) mandates transition to contribution-based contracts by 1 January 2028	OECD country profiles

Note: Entries use different metrics and are not directly comparable across institutional contexts. The UK entry constitutes the closest approximation to a quasi-experimental design; Chile and the Netherlands illustrate high-annuitization systems with distinct institutional structures; other entries represent cross-sectional observations or institutional time series.

The UK provides the strongest quasi-experimental evidence. The 2015 pension freedoms removed an effective annuitization requirement, and quarterly sales fell by approximately 75 percent (FCA, 2018). Annual sales rebounded to 89,600 contracts in 2024 as rates rose (ABI, 2025). TIAA-CREF documented a comparable decline within a single institutional setting: the share of first-time retirement income recipients selecting a life-contingent payout fell from 61 percent to 18 percent over 2000 to 2018 (Brown, Poterba, and Richardson, 2025), consistent with default-sensitive behavior as the institutional menu shifted. Singapore’s CPF LIFE achieved high compliance under mandatory status. Chile’s AFP system produces voluntary annuitization above 60 percent under structured choice architecture (Illanes and Padi, 2019), demonstrating that voluntary demand is not institutionally invariant. The Netherlands achieves near-universal annuitization through collective occupational arrangements rather than individual retail purchase. Taken together, these institutional contexts show that compliance is high when annuitization is mandated or embedded in institutional defaults, demand falls substantially when the mandate is removed, and take-up is low when annuitization is purely voluntary in retail contexts.

7. A Conditional Framework for Annuity Demand

The preceding sections review the principal channels proposed to explain low voluntary annuity demand. Table 3 inventories the channel-level evidence; Table 4 reorganizes that evidence into a conditional framework. The framework asks which channels bind for which households, which products and institutional settings are involved, and where the welfare-relevant question remains open.

The classifications use expected utility with empirically calibrated parameters as the welfare benchmark. Behavioral findings help explain observed choices, but loss aversion and framing effects are descriptive regularities rather than settled welfare criteria (Bernheim and Rangel, 2009). The labels follow explicit criteria. A cell is **well-explained** when at least two demand-reducing channels apply quantitatively to the modal household in that cell and at least one calibrated lifecycle model in the literature matches observed low participation for households of that type. A cell is **contested** when at least two credible calibrated models reach opposing welfare conclusions for the modal household, or when calibrated welfare-relevant magnitudes vary substantially across reasonable parameterizations. A cell is **under-examined** when product-specific welfare evidence for the household type is sparse or has not been integrated with household-heterogeneity models in the canonical puzzle literature. The classifications separate positive explanation from welfare judgment: a well-explained cell may still contain individual households who would benefit from annuitization, and welfare gains may be larger under improved product menus.

Table 3. Structured Inventory of Key Studies on the Annuity Puzzle

Study	Channel	Method	Key Finding	Effect on Predicted Annuitization
Yaari (1965)	Baseline	Analytical lifecycle model	Under EU, no bequests, and fair pricing, full annuitization is optimal	100% annuitization (theoretical benchmark)
Davidoff, Brown, and Diamond (2005)	Generalization / market completeness	Analytical model with relaxed preference assumptions	Full annuitization holds under weaker utility assumptions but requires complete markets; partial result survives under incomplete markets	100% under complete markets; potentially much lower under incomplete markets
Mitchell, Poterba, Warshawsky, and Brown (1999)	Supply-side pricing	Actuarial valuation of U.S. annuity contracts	Money's worth ratio of 0.80-0.85 for population-average mortality; 0.90-0.94 for annuitant mortality	Reduced; 15-20% load considerably diminishes welfare gains
Mitchell et al. (1999); Finkelstein and Poterba (2002, 2004)	Adverse selection	Actuarial decomposition; empirical analysis of UK market	Selection accounts for approximately 11 cents/dollar (roughly half of total pricing shortfall) for 65-year-old males; selection magnitude larger in voluntary than compulsory markets	Reduced for average-mortality individuals; selection raises effective price
Dushi and Webb (2004)	Pre-existing annuitization	Calibrated lifecycle model with SS and DB pensions	Accounting for pre-existing annuity income, optimal annuitization age ranges from 73 to 82 for couples; some should never annuitize	Sharply reduced; optimal timing delayed or eliminated
Hu and Scott (2007)	Prospect theory	Analytical model under CPT with Tversky-Kahneman parameters	Reservation prices fall below 1 under CPT with baseline parameterization and narrow framing; result sensitive to reference point and loss aversion coefficient	Predicted demand zero under baseline narrow-framing parameters; sensitivity to framing acknowledged
Brown, Kling, Mullainathan, and Wrobel (2008)	Framing effects	Survey experiment	72% prefer annuities in consumption frame vs. 21% in investment frame; 51-percentage-point swing from framing alone	Demand is frame-dependent, violating EU frame invariance
De Nardi, French, and Jones (2010)	Health expenditure risk	Structural lifecycle model with medical expense shocks	Medical expense risk drives elderly precautionary saving; liquid wealth needed for health shocks	Reduced; precautionary motive conflicts with annuitization
Ameriks, Caplin, Laufer, and Van Nieuwerburgh (2011)	Public care aversion	Strategic survey design separating bequest from care motives	Both bequest motives and public care aversion are economically significant; Medicaid aversion widespread	Reduced; households retain liquid wealth to self-fund LTC and avoid Medicaid
Lockwood (2012)	Bequest motives (calibration)	Calibrated lifecycle model with and without bequests	WTP collapses with bequest motives (25.3% to 3.7% of wealth for fair annuities); predicted ownership falls from 61% to 4-5% with bequests plus fixed purchase cost	Quantitative match on extensive margin requires bequest motives plus purchase frictions
Pashchenko (2013)	Multiple (pre-annuitization, bequests, housing, minimum purchase)	Structural lifecycle model with sequential decomposition	Full model predicts ~20% participation vs ~5% in data; tightened institutional specification reduces to under 8% without new channels	Substantially reduced but baseline model overpredicts

Study	Channel	Method	Key Finding	Effect on Predicted Annuity
Peijnenburg, Nijman, and Werker (2016)	Multiple (incomplete markets, background risk, bequests, default risk)	Calibrated lifecycle model	Full annuitization remains approximately optimal for reasonable parameter values; agents can save out of annuity income	Not reduced within single-model framework for reasonable parameters
Reichling and Smetters (2015)	Stochastic mortality and correlated health costs	Analytical and calibrated model	When mortality is stochastic and correlated with medical expenses, optimal annuity demand can be zero or negative	Zero or negative under calibrated health-mortality-cost correlation
Inkmann, Lopes, and Michaelides (2011)	Bequest motives, pre-existing income, preference parameters	Calibrated lifecycle model with UK data	Reasonable preference parameters predict annuity demand levels comparable to the data	Largely explained; reasonable parameters match observed low participation
O'Dea and Sturrock (2023)	Survival pessimism	Calibrated lifecycle model with ELSA subjective beliefs	Individuals underestimate survival by 15-30 pp; pessimism markedly reduces perceived annuity value	Reduced; survival pessimism accounts for significant non-annuitization within EU framework

Note: The “Effect on Predicted Annuity” column summarizes the direction and approximate magnitude of each channel’s effect relative to the Yaari benchmark of 100 percent annuitization. Rows are grouped approximately by channel: pricing and selection; pre-annuitization; behavioral and prospect theory; health and mortality; multi-channel structural models; and survival pessimism. Studies that reach contrasting welfare conclusions (Pashchenko, Peijnenburg et al., Inkmann et al.) anchor the contested cell in Table 4.

7.1 Retail single-premium immediate annuities

For most households, low SPIA take-up is well-explained by existing channels. Lower-wealth retirees hold much of their total wealth in annuitized Social Security and pension wealth: Lockwood (2012) estimates approximately 75 percent in the bottom quartile and 67 percent at the median. Among married households, intra-household longevity-risk pooling further reduces the incremental value of market annuitization; Kotlikoff and Spivak (1981) estimate that couples can capture approximately 70 percent of the welfare gains from fair annuity markets through informal pooling, and Brown and Poterba (2000) find lower utility gains from annuitization for married couples than for singles. At the top of the wealth distribution, couples with children combine bequest motives, housing wealth, intra-household pooling, and tax treatment. In Lockwood’s (2012) lifecycle calibration, the strong bequest parameterization of Ameriks et al. (2011) suffices to drive voluntary annuitization to zero, and the additional spousal-pooling channel reinforces this conclusion for couples.

The contested SPIA cell is the high-wealth, single, childless retiree in good health. In this case, the structural channels apply with reduced force: pre-annuitization is lower in the top decile, bequest motives are less prevalent though not absent, and expected longevity is higher. Pashchenko’s (2013) model predicts roughly 20 percent participation against about 5 percent in the data, although her tightened institutional specification reduces predicted participation below 8 percent without adding new channels. Peijnenburg, Nijman, and Werker (2016) find full annuitization approximately optimal when agents can rebuild liquid balances by saving out of annuity income, though their European calibration assumes negligible out-of-pocket health costs. Inkmann, Lopes, and Michaelides (2011), using UK data, find that reasonable preference parameters match observed low participation, with bequest motives and pre-existing pension income as key drivers. The disagreement across these three calibrations reflects specific modeling choices rather than fundamental theoretical conflict: Pashchenko incorporates US institutional frictions (Medicaid spend-down rules and minimum-purchase requirements) absent from Peijnenburg et al., who in turn allow agents to save out of annuity income while assuming negligible European out-of-pocket health costs; Inkmann et al. estimate preference parameters from UK household data in which substantial defined-benefit pension income operates as pre-annuitization. Each calibration choice is defensible on its own terms but leads to different welfare conclusions for the modal high-wealth childless single. This cell is where behavioral frictions, correlated health-mortality risk, and institutional calibration choices most directly determine whether low demand reflects welfare loss.

7.2 Deferred income annuities and QLACs

The channels that explain low SPIA demand apply less strongly to deferred income annuities (DIAs) and QLACs. DIA payouts supplement Social Security at advanced ages rather than duplicating it; the smaller upfront premium leaves more non-annuitized assets liquid; and the household retains the unannuitized portion of wealth until DIA payouts begin, typically at age 80 or 85. The SECURE 2.0 Act of 2022 raised the QLAC contribution limit to \$200,000, indexed for inflation (\$210,000 for 2025-2026), removing a previously binding constraint for many households (Public Law 117-328, §202; IRS Notice 2024-80).

Two frictions remain. First, inflation risk is more severe for nominal deferred products: a DIA purchased at 65 with payments beginning at 85 loses approximately 45 percent of real value before the first payment at 3 percent annual inflation. Second, money’s worth ratios for nominal DIAs are typically lower than for immediate annuities, though the gap is sensitive to product structure and mortality assumptions. Wettstein, Munnell, Hou, and Gok (2021) report that nominal DIA money’s worth ratios ranged across recent years from roughly 0.4 to 0.8 per premium dollar, while Poterba and Solomon (2021, revised 2025) estimate that for a DIA purchased at 65 with payouts beginning at 75, the expected present discounted value reached approximately 73 cents per premium dollar at population mortality and close to one dollar at annuitant mortality in May 2024. Yet near-zero take-up remains difficult to explain by these channels alone. Behavioral mechanisms are plausible: loss aversion against an immediate premium for remote contingent income, survival pessimism (O’Dea and Sturrock, 2023), and the cognitive burden of evaluating a decades-long irreversible commitment. Hu and Scott (2007) note that deferred annuities can be attractive under cumulative prospect theory for late starting ages. Across household types, DIA and QLAC welfare remains under-examined.

7.3 Pension payout choices

Pension payout choices occupy a contested middle ground between retail choice and institutional design. Brown, Poterba, and Richardson (2025) document that among TIAA participants, the share of first-time retirement income recipients choosing a life-contingent payout fell from 61 percent to 18 percent between 2000 and 2018; among those waiting until required minimum distribution age, the share fell from 41 percent to about 6 percent. This occurred in a relatively favorable institutional environment with trusted products and financially educated participants. The most visible change was not household preferences but the payout environment: systematic withdrawals, lump-sum access, and RMDs became more salient competing defaults. The welfare question is contested because the same evidence can be read as revealed preference under expanded choice or as default-sensitive behavior with possible welfare losses.

7.4 Mandatory and default regimes

Cross-institutional evidence shows that annuitization is highly responsive to policy design. In the UK, the 2015 pension freedoms removed an effective annuitization requirement, and quarterly annuity sales fell by approximately 75 percent, from 74,092 to 18,243 (FCA, 2018). Annual sales later rebounded to 89,600 contracts in 2024 as rates rose (ABI, 2025), indicating that pricing matters, but not eliminating the role of institutional choice architecture. Singapore’s CPF LIFE achieved high compliance after becoming mandatory (Fong, Mitchell, and Koh, 2011). Chile provides a useful counterweight: voluntary annuitization above 60 percent under structured AFP choice architecture suggests that voluntary demand is not institutionally invariant (Illanes and Padi, 2019). The Netherlands achieves near-universal annuitization through collective occupational arrangements rather than retail purchase.

These cases establish that institutions shape outcomes. They do not by themselves

establish welfare. Mandates and defaults may improve welfare when households underweight longevity risk, but may reduce welfare when optimal annuitization varies sharply by health, bequest motives, wealth, and liquidity needs. The mandatory/default regime cells are therefore contested.

Social Security claiming also belongs in the framework because it is an implicit annuitization decision. Delaying claims from age 62 to 70 yields delayed retirement credits, inflation indexation, and survivor benefits on terms unavailable in private markets. The standard annuity-puzzle literature usually treats Social Security as background income rather than a choice variable, so Table 4 classifies this row as under-examined. Section 8 develops the research agenda for this margin.

7.5 Framework Summary

Table 4 summarizes the conditional framework. Low voluntary SPIA demand is well-explained for lower-wealth households, median married households, and high-wealth couples with children. It remains contested for high-wealth, single, childless, healthy retirees. The principal under-examined margins are deferred income annuities, QLACs, and Social Security claiming, while pension payout choices and mandatory/default regimes remain contested because welfare depends on the design of defaults and the heterogeneity of affected households.

Table 4. Conditional framework: when is low annuity demand well-explained by existing channels, and where does it remain welfare-relevant?

Product / setting	Low-wealth single	Median married, DB-covered	High-wealth single, no children, healthy	High-wealth couple with children
Retail SPIA	Well-explained: pre-annuitization (Lockwood 2012; Dushi & Webb 2004)	Well-explained: pre-annuitization + spousal pooling + bequests (Kotlikoff & Spivak 1981; Brown & Poterba 2000)	Contested: Pashchenko (2013) overpredicts; Inkmann et al. (2011) match UK data; Peijnenburg et al. (2016) find full annuitization optimal	Well-explained: bequests + housing + tax asymmetry (Lockwood 2012; Ameriks et al. 2011)
DIA / QLAC	Under-examined: liquidity-constrained households unlikely to commit to long-deferred premiums	Under-examined: existing pre-annuitization already covers advanced-age income	Under-examined: strongest theoretical target given no bequest motive, ample liquidity, and long expected lifespan	Under-examined: bequest motive and joint mortality reduce relative attractiveness
Pension payout choice	Contested: lump sum often a small share of total wealth; default-sensitive (Brown, Poterba & Richardson 2025)	Contested: central group affected by default architecture	Contested: lump-sum optionality is more attractive given investment opportunities	Contested: bequest considerations favor lump sums even under annuity defaults
Social Security delay	Under-examined: bridge-asset constraint may bind; early claiming common (Shoven & Slavov 2014)	Under-examined: survivor benefits make primary-earner delay particularly valuable for couples	Under-examined: strong actuarial advantage to delay under above-average longevity	Under-examined: primary-earner delay maximizes joint-and-survivor present value
Mandatory / default regimes	Contested: mandates may improve welfare for low pre-annuitization; lock-in cost from inflexibility (FCA 2018; Fong, Mitchell & Koh 2011)	Contested: pre-annuitization makes additional mandate less critical	Contested: mandate-driven annuitization may reduce welfare for households with strong autonomy preferences	Contested: bequest preferences in tension with mandated annuitization

Note: “Well-explained” indicates that established channels in the calibrated literature account for observed low demand for the modal household in that cell. The marginal welfare gain from additional voluntary annuitization is typically modest in well-explained cells under current pricing and product menus, but this does not preclude individual households from benefiting and welfare gains may be larger under improved pricing or product design. “Contested” indicates that calibrated models or empirical estimates disagree on welfare-relevant magnitudes for the modal household. “Under-examined” indicates that the welfare question remains largely unaddressed within the canonical puzzle literature. Pre-retiree households aged 60-65 are the relevant population for QLAC purchase decisions within the DIA row. Classifications are conditional on current cohort characteristics — particularly substantial defined-benefit pension coverage — and several cells classified as well-explained should migrate toward contested or under-examined for future cohorts entering retirement with negligible DB coverage (Section 8.4).

The framework yields the research agenda developed in Section 8.

8. Remaining Questions and Directions for Future Research

The research agenda follows directly from the contested and under-examined cells in Table 4.

8.1 The Deferred Income Annuity Puzzle

The strongest residual challenge concerns deferred income annuities, including QLACs. These products preserve liquidity until advanced ages, use smaller premiums than SPIAs, and insure the segment of longevity risk most relevant to outliving assets. Because they directly attenuate the bequest and liquidity objections, near-zero demand is harder to explain through the standard SPIA channels. The open question is whether low DIA demand reflects welfare-relevant behavioral frictions, unattractive product design, or both.

Three issues deserve direct study. First, nominal deferred products expose households to severe inflation risk before payments begin. Second, reported money's worth ratios for DIAs are low in nominal terms (Wettstein et al., 2021), even if their insurance value may be higher on a utility basis. Third, survival pessimism and loss aversion may be especially strong for products whose payoff is remote and contingent on reaching advanced ages. DIA and QLAC experiments that vary inflation protection, default presentation, and purchase age would provide a sharp test of the behavioral-welfare question.

8.2 Heterogeneity in Optimal Annuitization

The most important SPIA research target is the high-wealth, healthy, childless retiree with low bequest motives and limited pre-annuitized income. Childless individuals account for approximately 16 percent of Americans aged 65-74 (U.S. Census Bureau, 2021), and roughly 25 percent of elderly singles are estimated to have no bequest motive (Kopczuk and Lupton, 2007). The intersection of high wealth, good health, low bequest motives, and limited pre-annuitization is empirically meaningful even if the joint distribution has not been precisely estimated. This is the group for whom retail annuitization may plausibly improve welfare under better pricing, inflation protection, or institutional pooling.

Future work should estimate heterogeneous welfare gains rather than average demand effects. The relevant empirical question is not whether annuities are valuable on average, but which households have positive marginal welfare gains from additional annuitization under realistic product menus.

8.3 Social Security Delay as Implicit Annuitization

Social Security delay is the most favorably priced annuitization mechanism available to many American retirees. Delaying benefits from age 62 to 70 increases monthly benefits through early-claiming reductions and delayed retirement credits, producing an inflation-indexed, survivor-benefit-eligible annuity backed by the federal government. For households with

bridge assets and average or better life expectancy, delay can dominate private annuitization (Shoven and Slavov, 2014).

The constraint is heterogeneity. Lower-income households may lack bridge liquidity or may face shorter life expectancy; higher-income households may benefit more from delay but also have more private substitutes. The annuity-puzzle literature typically treats Social Security as background income rather than a choice variable. Integrating optimal claiming into annuity-demand models would make the welfare comparison product-specific rather than categorical.

8.4 Cohort Dependence and the Migration of “Well-Explained” Cells

The pre-annuitization channel that drives the “well-explained” classification across much of Table 4 rests on cohorts with substantial defined-benefit pension coverage. As private-sector DB coverage continues to decline (currently roughly 14 percent of private-industry workers have access to a DB plan; Bureau of Labor Statistics 2025), the pre-annuitization share for future retirees will rest more heavily on Social Security alone. Several cells currently classified as “well-explained” should migrate toward “contested” or “under-examined” for cohorts entering retirement with negligible private DB coverage. The framework should be read as describing the current population of retirees rather than a stable steady state; the empirical and welfare questions will shift as the DC-dominant cohort ages into retirement.

A complementary substitute mechanism deserves explicit mention: housing wealth is the largest non-annuitized asset for the median retiree, and downsizing or reverse-mortgage arrangements function as informal longevity-management tools that compete with formal annuitization. The framework treats housing wealth as a tax-asymmetry-favored bequest vehicle, but it also serves as a substitute for additional voluntary annuitization for households that prefer to draw against home equity rather than convert financial wealth to an irrevocable income stream.

8.5 Directions for Future Research

Four additional directions follow. First, structural models jointly incorporating pre-existing annuitization, bequest motives, health risk, stochastic mortality, and realistic pricing are needed. The disagreement between Pashchenko (2013), Peijnenburg et al. (2016), and Inkmann et al. (2011) on the contested SPIA cell would be advanced by models allowing these channels to operate simultaneously.

Second, the defaults and choice-architecture literature (Carroll et al., 2009; Beshears et al., 2014) has direct implications for pension payout architecture. If the UK post-2015 experience and the TIAA-CREF decline reflect default effects rather than informed preference, default designs may prove more powerful levers than information provision or tax incentives, though the welfare and ethical implications of defaulting households into irrevocable lifetime-income products require separate attention.

Third, modern longevity-pooling designs warrant investigation. Milevsky and Salisbury (2015) analyze optimal retirement income tontines, pooled structures that distribute mortality credits without insurer guarantees, reducing insurer-guarantee exposure and potentially

lowering costs. Whether such designs would attenuate the liquidity, bequest, and counterparty-risk objections that suppress demand for conventional annuities is an open design question; they also introduce payout volatility and regulatory challenges that remain unresolved.

Fourth, data infrastructure that distinguishes among annuity product types (SPIA, DIA, GLWB, FIA, MYGA) is needed. Household surveys such as the HRS and SCF do not consistently distinguish among these product types, obscuring the product-specific welfare questions the framework identifies (Brown, 2007).

Two further directions deserve mention. The unitary-household treatment of annuitization in Kotlikoff and Spivak (1981) and most of the lifecycle literature abstracts from intra-household bargaining, gender mortality asymmetries, widow and widower transitions, and the rising prevalence of late-life marital dissolution. Each of these introduces a wedge between the household-level welfare object and individual outcomes. Models that relax the unitary assumption are likely to alter welfare conclusions for the median-married cell of Table 4. Relatedly, the framework’s “well-explained” verdicts are conditional on expected-utility welfare benchmarking. Under behavioral welfare frameworks that take loss aversion or framing-dependent preferences as welfare-relevant rather than as descriptive deviations (Bernheim and Rangel, 2009; subsequent work), several currently well-explained cells would migrate to contested.

Several boundary conditions warrant explicit acknowledgment. Pre-annuitization estimates depend on cohorts with substantial DB coverage and will shift as private DB rolls off; calibrated welfare predictions depend on bequest parameterizations that are empirically defensible but not unique; welfare conclusions depend on the expected-utility welfare benchmark; and alternative behavioral welfare frameworks (Bernheim and Rangel, 2009) may yield different assessments for the subpopulation identified in Section 8.2.

9. Conclusion

The annuity puzzle has occupied household finance researchers for six decades. The accumulated evidence supports a conditional reading of the gap between Yaari’s full-annuitization prediction and observed take-up of roughly 3 to 6 percent in the US and UK. For many households, low SPIA demand is well-explained by pre-existing annuitization, bequest motives, intra-household pooling, tax treatment, and pricing loads operating jointly. The clearest retail-SPIA contested cell is the high-wealth, single, childless household in good health, where Pashchenko (2013), Peijnenburg, Nijman, and Werker (2016), and Inkmann, Lopes, and Michaelides (2011) reach contrasting conclusions.

The welfare-relevant questions that remain open or under-examined fall in three categories: deferred income annuities and QLACs, where structural channels are weak and demand approaches zero across household types; Social Security claiming and pension payout defaults, where the welfare question has largely been treated outside the canonical puzzle tradition; and the design of mandatory or default annuitization regimes, where institutional choice architecture demonstrably shapes outcomes but the welfare implications across heterogeneous households remain contested.

Much of the gap between Yaari’s full-annuitization prediction and observed take-up can

be accounted for by the channels reviewed here, particularly for households with substantial pre-existing annuitization, operative bequest motives, or intra-household risk sharing. The productive research agenda lies in the cells where these channels apply with less force or where the literature has not yet engaged, such as heterogeneous welfare under improved pricing and product design, the architecture of pension payout defaults, deferred income annuities, and Social Security claiming as a household choice variable.

References

- Agnew, Julie R., Lisa R. Anderson, Jeffrey R. Gerlach, and Lisa R. Szykman. 2008. “Who Chooses Annuities? An Experimental Investigation of the Role of Gender, Framing, and Defaults.” *American Economic Review* 98 (2): 418–422. <https://doi.org/10.1257/aer.98.2.418>
- Alexandrova, Maria, and Nadine Gatzert. 2019. “What Do We Know About Annuitization Decisions?” *Risk Management and Insurance Review* 22 (1): 57–100. <https://doi.org/10.1111/rmir.12115>
- Ameriks, John, Andrew Caplin, Steven Laufer, and Stijn Van Nieuwerburgh. 2011. “The Joy of Giving or Assisted Living? Using Strategic Surveys to Separate Public Care Aversion from Bequest Motives.” *Journal of Finance* 66 (2): 519–561. <https://doi.org/10.1111/j.1540-6261.2010.01641.x>
- Association of British Insurers (ABI). 2025. “Another Post-Pension Freedoms Record for Annuity Sales.” Press release, February 12, 2025. London: ABI. <https://www.abi.org.uk/news/news-articles/2025/2/another-post-pension-freedoms-record-for-annuity-sales/>
- Australian Treasury. 2023. “Retirement Phase of Superannuation: Discussion Paper.” Canberra: Commonwealth of Australia. <https://treasury.gov.au/consultation/c2023-441613>
- Benartzi, Shlomo, Alessandro Previtero, and Richard H. Thaler. 2011. “Annuitization Puzzles.” *Journal of Economic Perspectives* 25 (4): 143–164. <https://doi.org/10.1257/jep.25.4.143>
- Bernheim, B. Douglas. 1991. “How Strong Are Bequest Motives? Evidence Based on Estimates of the Demand for Life Insurance and Annuities.” *Journal of Political Economy* 99 (5): 899–927. <https://doi.org/10.1086/261783>
- Bernheim, B. Douglas, and Antonio Rangel. 2009. “Beyond Revealed Preference: Choice-Theoretic Foundations for Behavioral Welfare Economics.” *Quarterly Journal of Economics* 124 (1): 51–104. <https://doi.org/10.1162/qjec.2009.124.1.51>
- Beshears, John, James J. Choi, David Laibson, Brigitte C. Madrian, and Stephen P. Zeldes. 2014. “What Makes Annuitization More Appealing?” *Journal of Public Economics* 116: 2–16. <https://doi.org/10.1016/j.jpubeco.2013.05.007>
- Bommier, Antoine, and Francois Le Grand. 2014. “Too Risk Averse to Purchase Insurance? A Theoretical Glance at the Annuity Puzzle.” *Journal of Risk and Uncertainty* 48 (2): 135–166. <https://doi.org/10.1007/s11166-014-9190-3>
- Brown, Jeffrey R. 2007. “Rational and Behavioral Perspectives on the Role of Annuities in Retirement Planning.” NBER Working Paper No. 13537. <https://doi.org/10.3386/w13537>
- Brown, Jeffrey R., and James M. Poterba. 2000. “Joint Life Annuities and Annuity Demand by Married Couples.” *Journal of Risk and Insurance* 67 (4): 527–553. <https://doi.org/10.2307/253849>

- Brown, Jeffrey R., Gopi Shah Goda, and Kathleen McGarry. 2016. “Heterogeneity in State-Dependent Utility: Evidence from Strategic Surveys.” *Economic Inquiry* 54 (2): 847–861. <https://doi.org/10.1111/ecin.12313>
- Brown, Jeffrey R., Jeffrey R. Kling, Sendhil Mullainathan, and Marian V. Wrobel. 2008. “Why Don’t People Insure Late-Life Consumption? A Framing Explanation of the Under-Annuity Puzzle.” *American Economic Review* 98 (2): 304–309. <https://doi.org/10.1257/aer.98.2.304>
- Brown, Jeffrey R., Olivia S. Mitchell, and James M. Poterba. 2002. “Mortality Risk, Inflation Risk, and Annuity Products.” In *Innovations in Retirement Financing*, edited by Olivia S. Mitchell, Zvi Bodie, P. Brett Hammond, and Stephen Zeldes, 175–197. Philadelphia: University of Pennsylvania Press.
- Brown, Jeffrey R., James M. Poterba, and David P. Richardson. 2025. “Trends in Retirement and Retirement Income Choices by TIAA Participants: 2000–2018.” *Journal of Pension Economics & Finance* 24 (1): 47–68. <https://doi.org/10.1017/S1474747223000070>
- Bureau of Labor Statistics. 2025. “Employee Benefits in the United States, March 2025.” News Release USDL-25-1464, September 25, 2025, Table 1. Washington, DC: U.S. Department of Labor. https://www.bls.gov/news.release/archives/ebs2_09252025.htm
- Carlin, Bruce I. 2009. “Strategic Price Complexity in Retail Financial Markets.” *Journal of Financial Economics* 91 (3): 278–287. <https://doi.org/10.1016/j.jfineco.2008.05.002>
- Carroll, Gabriel D., James J. Choi, David Laibson, Brigitte C. Madrian, and Andrew Metrick. 2009. “Optimal Defaults and Active Decisions.” *Quarterly Journal of Economics* 124 (4): 1639–1674. <https://doi.org/10.1162/qjec.2009.124.4.1639>
- Central Provident Fund Board (CPF Board). n.d. “Retirement Statistics: CPF LIFE.” Singapore: CPF Board. <https://www.cpf.gov.sg/member/infocenter/reports-and-statistics/cpf-statistics/retirement-statistics>
- Davidoff, Thomas, Jeffrey R. Brown, and Peter A. Diamond. 2005. “Annuities and Individual Welfare.” *American Economic Review* 95 (5): 1573–1590. <https://doi.org/10.1257/000282805775014281>
- De Nardi, Mariacristina, Eric French, and John B. Jones. 2010. “Why Do the Elderly Save? The Role of Medical Expenses.” *Journal of Political Economy* 118 (1): 39–75. <https://doi.org/10.1086/651674>
- Dushi, Irena, and Anthony Webb. 2004. “Household Annuity Decisions: Simulations and Empirical Analyses.” *Journal of Pension Economics & Finance* 3 (2): 109–143. <https://doi.org/10.1017/S1474747204001696>
- Financial Conduct Authority (FCA). 2018. *Retirement Outcomes Review: Final Report*. MS16/1.3. London: Financial Conduct Authority. <https://www.fca.org.uk/publications/market-studies/retirement-outcomes-review>
- Financial Conduct Authority (FCA). 2019. *Retirement Income Market Data 2018/19*. London:

- Financial Conduct Authority. <https://www.fca.org.uk/data/retirement-income-market-data-2018-19>
- Finkelstein, Amy, Erzo F. P. Luttmer, and Matthew J. Notowidigdo. 2013. “What Good Is Wealth Without Health? The Effect of Health on the Marginal Utility of Consumption.” *Journal of the European Economic Association* 11 (S1): 221–258. <https://doi.org/10.1111/j.1542-4774.2012.01101.x>
- Finkelstein, Amy, and James Poterba. 2002. “Selection Effects in the United Kingdom Individual Annuities Market.” *Economic Journal* 112 (476): 28–50. <https://doi.org/10.1111/1468-0297.00672>
- Finkelstein, Amy, and James Poterba. 2004. “Adverse Selection in Insurance Markets: Policyholder Evidence from the U.K. Annuity Market.” *Journal of Political Economy* 112 (1): 183–208. <https://doi.org/10.1086/379936>
- Fong, Joelle H. Y., Olivia S. Mitchell, and Benedict S. K. Koh. 2011. “Longevity Risk Management in Singapore’s National Pension System.” *Journal of Risk and Insurance* 78 (4): 961–982. <https://doi.org/10.1111/j.1539-6975.2010.01401.x>
- Genworth. 2024. *Cost of Care Survey 2024*. Richmond, VA: Genworth Financial / CareScout. <https://www.genworth.com/aging-and-you/finances/cost-of-care.html>
- Hamilton, W. D. 1964. “The Genetical Evolution of Social Behaviour. I.” *Journal of Theoretical Biology* 7 (1): 1–16. [https://doi.org/10.1016/0022-5193\(64\)90038-4](https://doi.org/10.1016/0022-5193(64)90038-4)
- Hooper, Paul L., Michael Gurven, Jeffrey Winking, and Hillard S. Kaplan. 2015. “Inclusive Fitness and Differential Productivity across the Life Course Determine Intergenerational Transfers in a Small-Scale Human Society.” *Proceedings of the Royal Society B: Biological Sciences* 282 (1803): 20142808. <https://doi.org/10.1098/rspb.2014.2808>
- Hu, Wei-Yin, and Jason S. Scott. 2007. “Behavioral Obstacles in the Annuity Market.” *Financial Analysts Journal* 63 (6): 71–82. <https://doi.org/10.2469/faj.v63.n6.4928>
- Illanes, Gastón, and Manisha Padi. 2019. “Retirement Policy and Annuity Market Equilibria: Evidence from Chile.” NBER Working Paper No. 26285. <https://doi.org/10.3386/w26285>
- Inkmann, Joachim, Paula Lopes, and Alexander Michaelides. 2011. “How Deep Is the Annuity Market Participation Puzzle?” *Review of Financial Studies* 24 (1): 279–319. <https://doi.org/10.1093/rfs/hhq080>
- Internal Revenue Service. 2024. “Notice 2024-80: 2025 Amounts Relating to Retirement Plans and IRAs, as Adjusted for Changes in Cost-of-Living.” Washington, DC: Internal Revenue Service. <https://www.irs.gov/pub/irs-drop/n-24-80.pdf>
- Jones, John Bailey, Mariacristina De Nardi, Eric French, Rory McGee, and Justin Kirschner. 2018. “The Lifetime Medical Spending of Retirees.” *Federal Reserve Bank of Richmond Economic Quarterly* 104 (3): 103–135. https://www.richmondfed.org/publications/research/economic_q
- Kaplan, Hillard, Kim Hill, Jane Lancaster, and A. Magdalena Hurtado. 2000. “A Theory of Human Life History Evolution: Diet, Intelligence, and Longevity.” *Evolutionary*

- Anthropology* 9 (4): 156–185. [https://doi.org/10.1002/1520-6505\(2000\)9:4<156::AID-EVAN5>3.0.CO;2-7](https://doi.org/10.1002/1520-6505(2000)9:4<156::AID-EVAN5>3.0.CO;2-7)
- Kopczuk, Wojciech, and Joseph P. Lupton. 2007. “To Leave or Not to Leave: The Distribution of Bequest Motives.” *Review of Economic Studies* 74 (1): 207–235. <https://doi.org/10.1111/j.1467-937X.2007.00419.x>
- Kotlikoff, Laurence J., and Avia Spivak. 1981. “The Family as an Incomplete Annuities Market.” *Journal of Political Economy* 89 (2): 372–391. <https://doi.org/10.1086/260970>
- Laibson, David. 1997. “Golden Eggs and Hyperbolic Discounting.” *Quarterly Journal of Economics* 112 (2): 443–478. <https://doi.org/10.1162/003355397555253>
- Lambregts, Timo R., and Frederik T. Schut. 2020. “Displaced, Disliked and Misunderstood: A Systematic Review of the Reasons for Low Uptake of Long-Term Care Insurance and Life Annuities.” *Journal of the Economics of Ageing* 17: 100236. <https://doi.org/10.1016/j.jeoa.2020.100236>
- Lillard, Lee A., and Yoram Weiss. 1997. “Uncertain Health and Survival: Effects on End-of-Life Consumption.” *Journal of Business & Economic Statistics* 15 (2): 254–268. <https://doi.org/10.1080/07350015.1997.10524704>
- LIMRA. 2025. “2024 Retail Annuity Sales Grow 13% to a Record \$434.1 Billion.” News release, March 11, 2025 (updated March 17, 2025). Windsor, CT: LIMRA. [https://www.limra.com/en/newsroom/news-releases/2025/limra-2024-retail-annuity-sales-grow-12-to-a-record-\\$434.1-billion/](https://www.limra.com/en/newsroom/news-releases/2025/limra-2024-retail-annuity-sales-grow-12-to-a-record-$434.1-billion/)
- Lockwood, Lee M. 2012. “Bequest Motives and the Annuity Puzzle.” *Review of Economic Dynamics* 15 (2): 226–243. <https://doi.org/10.1016/j.red.2011.03.001>
- Lusardi, Annamaria, and Olivia S. Mitchell. 2011. “Financial Literacy around the World: An Overview.” *Journal of Pension Economics & Finance* 10 (4): 497–508. <https://doi.org/10.1017/S1474747211000448>
- Madrian, Brigitte C., and Dennis F. Shea. 2001. “The Power of Suggestion: Inertia in 401(k) Participation and Savings Behavior.” *Quarterly Journal of Economics* 116 (4): 1149–1187. <https://doi.org/10.1162/003355301753265543>
- Milevsky, Moshe A., and Thomas S. Salisbury. 2015. “Optimal Retirement Income Tontines.” *Insurance: Mathematics and Economics* 64: 91–105. <https://doi.org/10.1016/j.insmatheco.2015.05.002>
- Mitchell, Olivia S., James M. Poterba, Mark J. Warshawsky, and Jeffrey R. Brown. 1999. “New Evidence on the Money’s Worth of Individual Annuities.” *American Economic Review* 89 (5): 1299–1318. <https://doi.org/10.1257/aer.89.5.1299>
- National Organization of Life and Health Insurance Guaranty Associations (NOLHGA). n.d. “How You’re Protected.” Herndon, VA: NOLHGA. <https://nolhga.com/policyholders/how-youre-protected/>
- O’Dea, Cormac, and David Sturrock. 2023. “Survival Pessimism and the Demand for Annuities.” *Review of Economics and Statistics* 105 (2): 442–457.

https://doi.org/10.1162/rest_a_01048

- OECD. 2023. *Pensions at a Glance 2023: Country Profiles – Netherlands*. Paris: OECD Publishing. <https://www.oecd.org/els/public-pensions/PAG2023-country-profile-Netherlands.pdf>
- Pashchenko, Svetlana. 2013. “Accounting for Non-Annuity.” *Journal of Public Economics* 98: 53–67. <https://doi.org/10.1016/j.jpubeco.2012.11.005>
- Peijnenburg, Kim, Theo Nijman, and Bas J. M. Werker. 2016. “The Annuity Puzzle Remains a Puzzle.” *Journal of Economic Dynamics and Control* 70: 18–35. <https://doi.org/10.1016/j.jedc.2016.05.023>
- Poterba, James M., and Adam Solomon. 2021. “New Evidence on the Money’s Worth of Immediate and Deferred Individual Annuities.” NBER Working Paper No. 28557, revised October 2025. <https://doi.org/10.3386/w28557>
- Poterba, James M., Steven F. Venti, and David A. Wise. 2011. “The Composition and Drawdown of Wealth in Retirement.” *Journal of Economic Perspectives* 25 (4): 95–118. <https://doi.org/10.1257/jep.25.4.95>
- Reichling, Felix, and Kent Smetters. 2015. “Optimal Annuity with Stochastic Mortality and Correlated Medical Costs.” *American Economic Review* 105 (11): 3273–3320. <https://doi.org/10.1257/aer.20131584>
- Shoven, John B., and Sita N. Slavov. 2014. “Does It Pay to Delay Social Security?” *Journal of Pension Economics & Finance* 13 (2): 121–144. <https://doi.org/10.1017/S1474747213000309>
- U.S. Census Bureau. 2021. “First-Ever Census Bureau Report Highlights Growing Childless Older Adult Population.” Press Release CB21-156, August 31, 2021. Washington, DC: U.S. Census Bureau. <https://www.census.gov/newsroom/press-releases/2021/childless-older-adult-population.html>
- United States. 2022. *Consolidated Appropriations Act, 2023*. Public Law 117-328, Division T (SECURE 2.0 Act of 2022), §202. Washington, DC: U.S. Government Publishing Office. <https://www.congress.gov/117/plaws/publ328/PLAW-117publ328.pdf>
- Wettstein, Gal, Alicia H. Munnell, Wenliang Hou, and Nilufer Gok. 2021. “The Value of Annuities.” Center for Retirement Research at Boston College, Working Paper 2021-5. <https://doi.org/10.2139/ssrn.3797822>
- Yaari, Menahem E. 1965. “Uncertain Lifetime, Life Insurance, and the Theory of the Consumer.” *Review of Economic Studies* 32 (2): 137–150. <https://doi.org/10.2307/2296058>